Light-Driven Nanomotors with controllable 3D Motion

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Summary: We introduce a nanomotor capable of fully controllable 3D motion by combining two independent optical actuation channels. An azimuthally polarized Bessel beam couples to guided modes of a dielectric cylinder, enabling stable long-range optical pulling, while embedded asymmetric plasmonic dimers generate directional lateral forces under plane-wave illumination. Plasmonic dimers are arranged to avoid interaction with the Bessel beam to decouple the longitudinal and transversal motions, with simulations confirming robustness against misalignment and Brownian fluctuations. This design offers a practical, reconfigurable platform for advanced light-driven micro- and nanomachines.

ABSTRACT

We present a nanomotor architecture that achieves fully controllable three-dimensional motion by integrating two distinct optical actuation mechanisms within a single platform [1]. The design centers on a dielectric cylinder that simultaneously functions as the structural chassis and as an optical waveguide for azimuthally polarized Bessel beams [2,3]. When illuminated at moderate cone angles, the cylinder supports guided electromagnetic modes that reshape the scattering profile of the incident beam. This interaction leads to efficient optical pulling forces without relying on high numerical apertures or exotic material responses, thereby enabling longitudinal propulsion over micrometer distances.

To introduce independently tunable transverse motion, asymmetric plasmonic nanorod dimers are embedded within the cylinder [4,5]. Their orientation is chosen to suppress their response under Bessel-beam illumination, preserving the pulling mechanism; however, under linearly polarized plane waves, these dimers exhibit highly directional scattering that generates strong lateral optical forces. The independence of these two mechanisms allows the nanomotor to easily switch between pulling, pushing, and lateral movements simply by modifying the incident beam type or polarization. Furthermore, controlled, helicity-dependent rotations can be implemented, effectively achieving full independent 3-dimensional motion.

The pulling response is remarkably stable under Brownian motion at ambient temperature, according to the results of diffusion modeling fed by full-wave simulations. The restoring force and torque profiles produced by the Bessel-beam interaction allow the nanomotor to continue on its path in spite of off-axis displacements and rotations. This combination of long-range optical pulling, robust stability, and multi-degree-of-freedom controllability paves the way toward versatile, reconfigurable light-driven nanomachines for lab-on-a-chip systems, microfluidics, and advanced optical manipulation technologies.

Keywords: Nanotechnology, nanophotonics, plasmonics, optical forces, optical pulling.

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